Investigations on Higher-Order Infinitary Logic

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Abstract

Higher-order logic and infinitary logic are two extensions of first-order logic that allow greater expressivity. Both features have not been investigated together yet. In this paper, we define a higher-order infinitary logic, based on an extension of simple type theory. The resulting logic features higher-order quantifiers, infinite conjunctions and infinite disjunctions. We establish results at both the syntactic and the semantic level. We introduce a sound notion of model, and we show a strong version of completeness that entails the cut-elimination theorem for natural deduction. Moreover, we prove an extension of Barr's theorem, allowing us to constructivize classical proofs of a particular fragment of higher-order infinitary logic.

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1 Introduction

Infinitary logic is an extension of first-order logic that allows infinite long statements or proofs. This is typically achieved by featuring infinite sequences of conjunctions, disjunctions or quantifiers. Such expressive powers are useful when axiomatizing various mathematical concepts, but also when formalizing problems from graph theory or epistemic logic [30, 26]. Infinitary logic was first coined by Zermelo [37] for studying the foundations of set theory. It was formally established by Karp [19] and its properties were investigated in [3, 20]. Among those results is the cut-elimination theorem for infinitary logics [23, 34, 10, 36, 26].

Geometric formulas are particular infinitary formulas, built using finite conjunctions, infinite disjunctions and existential quantifiers. This is sufficient to formalize numerous mathematical structures, like groups, rings and fields [30]. Geometric formulas satisfy an interesting result of proof theory, known as Barr's theorem. It states that any geometric formula provable from a geometric theory using classical logic is also provable from the same theory using intuitionistic logic. Originally expressed in topos-theoretic terms [2], it has been proved in proof-theoretic terms [30, 26]. The original Barr's theorem also claims to remove the axiom of choice. Such an assertion has been investigated in [30].

Higher-order logic [4] proposes another route to extend first-order logic and boost its expressive powers. Rather than enhancing conjunction and disjunction, higher-order logic focuses on the universal and existential quantifiers. While first-order quantifiers range over individuals, higher-order quantifiers also have the ability to abstract over functions, predicates or formulas. It follows that higher-order logic is highly expressive [9].


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Both infinitary and higher-order features have not been investigated together yet. In this paper, our aim is to define and study a higher-order infinitary logic.

Contributions. The contribution of this paper is threefold. First, we extend simple type theory with a type, a constructor, and destructors for families of propositions indexed by an arbitrary countable set. We show that, in this setting, the β-reduction neither satisfies strong normalization nor confluence, but every well-typed term has a head normal form. Second, we define a higher-order infinitary logic, along with cut-free natural deduction inference rules. We define a sound notion of model and we prove a strong version of completeness that entails the cut-elimination theorem. Third, we show that Barr's theorem can be extended to higher-order geometric theories, allowing us to constructivize classical proofs of higher-order geometric formulas.

Outline. We present our extension of simple type theory with families of propositions in Section 2. We define higher-order infinitary logic and its natural deduction systems in Section 3. We prove the cut-elimination theorem in Section 4 and Barr's theorem in Section 5.

2 An Extension of Simple Type Theory

Higher-order logic is usually modeled using Church's simple type theory [4], where we have a type of propositions and a type of individuals (that are basic values). Here we want to model higher-order infinitary logic, that is to have an infinite conjunction and an infinite disjunction. In simple type theory, we cannot define functions taking infinitely many arguments. That is why we extend simple type theory with a new type for families of propositions indexed by an arbitrary countable set, along with a constructor and destructors.

▶ **Definition 1.** *Types and terms are defined inductively:*

$$\begin{array}{llll} \textit{Types} & \tau, \sigma & ::= & \iota \mid o \mid o_{\mathcal{I}} \mid \tau \to \sigma \\ \textit{Terms} & t, u, A_i, F & ::= & x \mid c \mid t\ u \mid \lambda x.\ t \mid \{i \mapsto A_i\}_{\mathcal{I}} \mid \pi_j\ F \end{array}$$

where ι is the type of individuals, o is the type of propositions, $o_{\mathcal{I}}$ is the type of families of propositions indexed by a countable set $\mathcal{I}$, x is a variable, c is a constant, $\{i \mapsto A_i\}_{\mathcal{I}}$ is the family of propositions $(A_i)_{i \in \mathcal{I}}$, and $\pi_j\ F$ is the projection on the j-th component of F.

▶ Remark 2. It is also possible to define our type theory with multiple base types $\iota_1, \ldots, \iota_n$ and multiple types of families of propositions $o_{\mathcal{I}_1}, \ldots, o_{\mathcal{I}_m}$.

The typing rules are the usual ones, extended with:
- if for every $i \in \mathcal{I}$ we have A_i of type o then $\{i \mapsto A_i\}_{\mathcal{I}}$ is of type $o_{\mathcal{I}}$,
- if F is of type $o_{\mathcal{I}}$ then $\pi_j\ F$ is of type o.

We write Λ_τ for the set of terms of type τ. We restrict terms so that they do not have infinitely many free variables. We write $t[x \leftarrow u]$ for the term t in which x has been substituted by u.

Computation is introduced in this type theory through the usual β-reduction rule and η-contraction rule, extended with rules characterizing the projections.

▶ **Definition 3.** *We consider the following rules:*
- *the β-reduction rule $(\lambda x.\ t)\ u \hookrightarrow t[x \leftarrow u]$,*
- *the projection rules $\pi_j\ \{i \mapsto A_i\}_{\mathcal{I}} \hookrightarrow A_j$,*
- *the η-contraction rule $\lambda x.\ t\ x \hookrightarrow t$, where x is not free in t,*
- *the rule $\{i \mapsto \pi_i\ F\}_{\mathcal{I}} \hookrightarrow F$.*

The reduction relation $\hookrightarrow$ is the smallest relation that contains these rules and such that:
- *if $t \hookrightarrow t'$, then $t\ u \hookrightarrow t'\ u$, $u\ t \hookrightarrow u\ t'$ and $\lambda x.\ t \hookrightarrow \lambda x.\ t'$.*
- *if $A_i \hookrightarrow B_i$ for every $i \in \mathcal{J}$, where $\mathcal{J} \neq \emptyset$ and $\mathcal{J} \subseteq I$, and $A_i = B_i$ otherwise, then $\{i \mapsto A_i\}_{\mathcal{I}} \hookrightarrow \{i \mapsto B_i\}_{\mathcal{I}}$,*
- *if $F \hookrightarrow F'$ then $\pi_j\ F \hookrightarrow \pi_j\ F'$.*

The relation $\equiv$ is the reflexive, symmetric and transitive closure of $\hookrightarrow$.

The $\beta\pi$-reduction $\hookrightarrow_{\beta\pi}$ and the $\beta\pi$-conversion $\equiv_{\beta\pi}$ are defined like $\hookrightarrow$ and $\equiv$, but only consider the β-reduction rule and the projection rules.

Properties. We have extended simple type theory with families of propositions. We have to check whether the intuition and expected results are confirmed.

▶ **Lemma 4** (Substitution). *Let t be a term of type τ with x a variable of type σ, and u be a term of type σ. Then $t[x \leftarrow u]$ is of type τ.*

Proof. By induction on t. ◀

▶ **Lemma 5** (Subject Reduction). *If t is of type τ and $t \hookrightarrow u$, then u is of type τ.*

Proof. By induction on the derivation of $t \hookrightarrow u$.
- Suppose that $(\lambda x.\ t')\ v \hookrightarrow t'[x \leftarrow v]$. We have x of type σ, t' of type τ and v of type σ. By Lemma 4, we derive that $t'[x \leftarrow v]$ is of type τ.
- Suppose that $\pi_j\ \{i \mapsto A_i\}_{\mathcal{I}} \hookrightarrow A_j$. We necessarily have $\pi_j\ \{i \mapsto A_i\}_{\mathcal{I}}$ of type o and thus A_j of type o.
- Suppose that $\{i \mapsto \pi_i\ F\}_{\mathcal{I}} \hookrightarrow F$. We necessarily have $\{i \mapsto \pi_i\ F\}_{\mathcal{I}}$ of type $o_{\mathcal{I}}$ and thus $\pi_i\ F$ of type o and F of type $o_{\mathcal{I}}$.
- Suppose that $A_i \hookrightarrow B_i$ for every $i \in \mathcal{J}$ (where $\mathcal{J} \neq \emptyset$ and $\mathcal{J} \subseteq I$) and $A_i = B_i$ otherwise. We have $\{i \mapsto A_i\}_{\mathcal{I}}$ of type $o_{\mathcal{I}}$, hence every A_i is of type o. Thus every B_i is of type o (by induction hypothesis when $i \in \mathcal{J}$ and by definition otherwise). It follows that $\{i \mapsto B_i\}_{\mathcal{I}}$ is of type $o_{\mathcal{I}}$.
- The other cases are proved similarly. ◀

In this extension of simple type theory, the $\beta\pi$-reduction relation is neither terminating nor confluent. For every $n \in \mathbb{N}^*$, let A_n be $P\ x \Rightarrow \ldots \Rightarrow P\ x$, where $P\ x$ occurs n times. Consider the term $(\lambda x.\ \{n \mapsto A_n\}_{\mathbb{N}^*})\ ((\lambda z.\ t)\ u)$. On the one hand, it reduces to $(\lambda x.\ \{n \mapsto A_n\}_{\mathbb{N}^*})\ t[z \leftarrow u]$ and then $\{n \mapsto A_n[x \leftarrow t[z \leftarrow u]]\}_{\mathbb{N}^*}$. On the other hand, it reduces to $\{n \mapsto A_n[x \leftarrow (\lambda z.\ t)\ u]\}_{\mathbb{N}^*}$. Each term $A_n[x \leftarrow (\lambda z.\ t)\ u]$ reduces to $A_n[x \leftarrow t[z \leftarrow u]]$ in n steps. But the family $\{n \mapsto A_n[x \leftarrow (\lambda z.\ t)\ u]\}_{\mathbb{N}^*}$ does not reduce to $\{n \mapsto A_n[x \leftarrow t[z \leftarrow u]]\}_{\mathbb{N}^*}$, as there will always be a proposition $A_n[x \leftarrow (\lambda z.\ t)\ u]$ that has not yet reached $A_n[x \leftarrow t[z \leftarrow u]]$. This is a counter-example of both strong normalization and confluence for the β-reduction. We can also build a counter-example of strong normalization and confluence for the $\beta\pi$-reduction, using $\pi_m\ \{n \mapsto B_n\}_{\mathbb{N}}$ instead of $((\lambda z.\ t)\ u)$ and B_m instead of $t[z \leftarrow u]$. We shall see, however, that well-typed terms have $\beta\pi$-head-normal forms.

▶ **Definition 6** ($\beta\pi$-Head Normal Forms). *$\beta\pi$-head normal forms are inductively defined by:*
- *an application $t\ u_1\ \ldots\ u_n$ where t is a variable or a constant and possibly $n = 0$,*
- *a projection $\pi_j\ (t\ u_1\ \ldots\ u_n)$ where t is a variable or a constant and possibly $n = 0$,*
- *an abstraction $\lambda x.\ t$ where t is a $\beta\pi$-head normal form,*
- *a family of propositions $\{i \mapsto A_i\}_{\mathcal{I}}$.*

A term has a $\beta\pi$-head normal form when it is a $\beta\pi$-head normal form or when it $\beta\pi$-reduces to a $\beta\pi$-head normal form.

▶ Remark 7. In the case for families of propositions $\{i \mapsto A_i\}_{\mathcal{I}}$, we do not require every A_i to be a $\beta\pi$-head normal form. Otherwise, we could create a counter-example in which each A_n reaches a $\beta\pi$-head normal form in n steps, so it would be impossible for the family of propositions $\{n \mapsto A_n\}_{\mathbb{N}}$ to reach a $\beta\pi$-head normal form in a finite number of steps.

We show that every well-typed term has a $\beta\pi$-head normal form. To do so, we adapt the well-known reducibility technique [33, 14].

▶ **Definition 8.** *Let τ be a type. The set of terms R_τ is defined by induction:*

- *When $\tau \in \{\iota, o\}$, a term t belongs to R_τ if and only if it is of type τ and has a $\beta\pi$-head normal form.*
- *When $\tau = o_{\mathcal{I}}$, a term t belongs to $R_{o_{\mathcal{I}}}$ if and only if it is of type $o_{\mathcal{I}}$, it has a $\beta\pi$-head normal form, and for every $i \in \mathcal{I}$, we have $(\pi_i\ t) \in R_o$.*
- *When $\tau = \tau_1 \to \tau_2$, a term t belongs to R_τ if and only if it is of type $\tau_1 \to \tau_2$, it has a $\beta\pi$-head normal form, and for every term $u \in R_{\tau_1}$, we have $(t\ u) \in R_{\tau_2}$.*

By definition, if $t \in R_\tau$ then t is of type τ and has a $\beta\pi$-head normal form.

▶ **Lemma 9.** *Let t, u be terms of type τ such that $t \hookrightarrow_{\beta\pi} u$. If $u \in R_\tau$ then $t \in R_\tau$.*

Proof. By induction on type τ. As $u \in R_\tau$, we know that u has a $\beta\pi$-head normal form, and thus t also has a $\beta\pi$-head normal form.

- If $\tau \in \{\iota, o\}$, we have $t \in R_\tau$ by definition.
- If $\tau = o_{\mathcal{I}}$, then for every $i \in \mathcal{I}$ we have $(\pi_i\ u) \in R_o$, and thus $(\pi_i\ t) \in R_o$. Hence $t \in R_{o_{\mathcal{I}}}$.
- If $\tau = \tau_1 \to \tau_2$, for every $v \in R_{\tau_1}$, we have $(u\ v) \in R_{\tau_2}$. Since $t\ v \hookrightarrow_{\beta\pi} u\ v$, we get $(t\ v) \in R_{\tau_2}$ by induction hypothesis. We conclude $t \in R_{\tau_1 \to \tau_2}$. ◀

▶ **Lemma 10.** *Let $t\ u_1\ \ldots\ u_n$ be a term of type τ, with t a variable or a constant and possibly $n = 0$. Then $(t\ u_1\ \ldots\ u_n) \in R_\tau$.*

Proof. By induction on type τ. The term $t\ u_1\ \ldots\ u_n$ is directly a $\beta\pi$-head normal form of type τ.

- If $\tau \in \{\iota, o\}$, then $(t\ u_1\ \ldots\ u_n) \in R_\tau$ by definition.
- If $\tau = o_{\mathcal{I}}$, then for every $i \in \mathcal{I}$ we have $\pi_i\ (t\ u_1\ \ldots\ u_n) \in R_o$ (as it is directly a $\beta\pi$-head normal form of type o), and thus $(t\ u_1\ \ldots\ u_n) \in R_{o_{\mathcal{I}}}$.
- If $\tau = \tau_1 \to \tau_2$, then for every $v \in R_{\tau_1}$ we have $(t\ u_1\ \ldots\ u_n\ v) \in R_{\tau_2}$ (by induction hypothesis), and thus $(t\ u_1\ \ldots\ u_n) \in R_{\tau_1 \to \tau_2}$. ◀

▶ **Theorem 11.** *Let t be a term of type τ and θ be a substitution mapping each variable of type τ' to an element of $R_{\tau'}$. We have $t\theta \in R_\tau$.*

Proof. We proceed by induction on t.

- If t is a variable, then we directly have $t\theta \in R_\tau$.
- If t is a constant, then $t\theta = t$ and we have $t \in R_\tau$ using Lemma 10.
- If t is an application $(u\ v)$, then u has type $\sigma \to \tau$ and v has type σ. By induction hypotheses, $u\theta \in R_{\sigma\to\tau}$ and $v\theta \in R_\sigma$. By definition of $R_{\sigma\to\tau}$, we have $t\theta = (u\theta\ v\theta) \in R_\tau$.
- If t is an abstraction $\lambda x.\ t'$, then $\tau = \tau_1 \to \tau_2$ and t' has type τ_2. We want to show that $t\theta \in R_\tau$, that is $(\lambda x.\ t'\theta) \in R_{\tau_1\to\tau_2}$. By induction hypothesis, $t'\theta \in R_{\tau_2}$. It follows that $t'\theta$ has a $\beta\pi$-head normal form and therefore that $(\lambda x.\ t'\theta)$ has a $\beta\pi$-head normal form. Let $u \in R_{\tau_1}$. We have $(\lambda x.\ t'\theta)\ u \hookrightarrow_{\beta\pi} t'[\theta, x \leftarrow u]$. We have $t'[\theta, x \leftarrow u] \in R_{\tau_2}$ by induction hypothesis. Using Lemma 9, we get $((\lambda x.\ t'\theta)\ u) \in R_{\tau_2}$.
- If t is a projection $\pi_j\ F$, then $\tau = o$ and F has type $o_{\mathcal{I}}$. By induction hypothesis, $F\theta \in R_{o_{\mathcal{I}}}$. It follows that $(\pi_j\ F\theta) \in R_o$, that is $(\pi_j\ F)\theta \in R_o$.

- Suppose that t is a family of propositions $\{i \mapsto A_i\}_{\mathcal{I}}$. By definition, $\{i \mapsto A_i\theta\}_{\mathcal{I}}$ is a $\beta\pi$-head normal form. By induction hypothesis, we have $A_j\theta \in R_o$ for every $j \in \mathcal{I}$, and thus $\pi_j\ \{i \mapsto A_i\theta\}_{\mathcal{I}} \in R_o$. Therefore $t\theta \in R_{o_{\mathcal{I}}}$. ◀

▶ **Corollary 12.** *Every well-typed term has a $\beta\pi$-head normal form.*

Proof. Let t be a term of type τ. We take the substitution θ that maps each variable x of type τ' to itself. We have $x \in R_{\tau'}$ by Lemma 10. We apply Theorem 11 and we get $t\theta = t \in R_\tau$. By definition of R_τ, t has a $\beta\pi$-head normal form. ◀

▶ Remark 13. Weak head normal forms can be defined like head normal forms, except that the condition on t in the abstraction case is dropped. From Corollary 12, we directly get that every well-typed term has a weak head normal form.

3 Higher-Order Infinitary Logic

Syntax. There are particular constants, called logical constants, for defining the connectives and quantifiers: contradiction $\bot$ of type o, implication $\Rightarrow$ of type $o \to o \to o$, quantifiers $\forall_\tau$ and $\exists_\tau$ of type $(\tau \to o) \to o$, infinite conjunction $\bigwedge_{\mathcal{I}}$ and infinite disjunction $\bigvee_{\mathcal{I}}$ of type $o_{\mathcal{I}} \to o$. For convenience, terms of the form $\forall_\tau\ (\lambda x.\ A)$ and $\exists_\tau\ (\lambda x.\ A)$ are simply abbreviated as $\forall x.A$ and $\exists x.A$. The negation is defined by $\neg A := A \Rightarrow \bot$ and the logical biconditional $\Leftrightarrow$ is defined by $A \Leftrightarrow B := (A \Rightarrow B) \wedge (B \Rightarrow A)$. Formulas are terms of type o.

▶ **Example 14.** Suppose that we want to model the interactions between researchers, represented by terms of type ι. We have several binary relations of type $\iota \to \iota \to o$, for example coauthors, colleagues or friends. We want to know whether two researchers x and y are related, no matter the number of intermediate researchers between them or the nature of the relations involved. Intuitively, we check if x and y are related in n steps, and we do this until we have found such a $n \in \mathbb{N}^*$. In our logic, we easily model this problem with

$$\bigvee_{\mathbb{N}^*} \{n \mapsto \exists z_0 \ldots \exists z_n . \exists R_1 \ldots \exists R_n .\ x = z_0 \wedge y = z_n \wedge R_1\ z_0\ z_1 \wedge \ldots \wedge R_n\ z_{n-1}\ z_n\}_{\mathbb{N}^*}$$

which features an infinite disjunction and higher-order quantifiers.

When taking $y =$ Erdos, this problem generalizes Erdős number to any kind of relations between researchers. We exactly obtain Erdős number when the relations R_i are coauthors.

Natural deduction. A context Γ is a finite sequence of formulas. We write $\Gamma \vdash A$ when A is derivable from Γ. We write $\mathsf{fv}(t_1, \ldots, t_n)$ for the set of free variables that occur in the terms $t_1, \ldots, t_n$. We define a cut-free natural deduction system following [6, 7, 13]. We introduce two restrictions of the turnstile $\vdash$, to respectively account for cut-free derivations ($\vdash^{\uparrow}$), and neutral cut-free derivations ($\vdash^{\downarrow}$). The cut-free natural deduction inference rules are given in Figure 1, where $\vdash^*$ is used to denote either $\vdash^{\downarrow}$, or $\vdash^{\uparrow}$. The Coerce rule turns neutral cut-free proofs into cut-free proofs. The natural deduction inference rules (with cuts) is obtained by ignoring the adornment of the turnstile – the Coerce rule becomes vacuously admissible with the plain turnstile relation.

The Conv rule allows us to consider formulas modulo the congruence $\equiv$, since we are in higher-order logic. Note that the Conv rule implicitly requires that B is of type o. The natural deduction rules for the infinite conjunction and disjunction generalize the usual natural deduction rules for the finite conjunction and disjunction. There are versions of the rules AndInf-E and OrInf-I for every $j \in \mathcal{I}$. Note that, due to the OrInf-E and AndInf-I inference rules, derivation trees possibly have an infinite width, but each branch

$$\frac{}{\Gamma, A, \Delta \vdash^{\downarrow} A}\ \textsc{Ax} \qquad \frac{\Gamma \vdash^{*} A \qquad A \equiv B}{\Gamma \vdash^{*} B}\ \textsc{Conv} \qquad \frac{\Gamma \vdash^{\downarrow} A}{\Gamma \vdash^{\uparrow} A}\ \textsc{Coerce}$$

$$\frac{\Gamma, A \vdash^{\uparrow} B}{\Gamma \vdash^{\uparrow} A \Rightarrow B}\ \textsc{Imp-I} \qquad \frac{\Gamma \vdash^{\downarrow} A \Rightarrow B \qquad \Gamma \vdash^{\uparrow} A}{\Gamma \vdash^{\downarrow} B}\ \textsc{Imp-E}$$

$$\frac{\Gamma \vdash^{\uparrow} \pi_i\ F \text{ for every } i \in \mathcal{I}}{\Gamma \vdash^{\uparrow} \bigwedge_{\mathcal{I}} F}\ \textsc{AndInf-I} \qquad \frac{\Gamma \vdash^{\downarrow} \bigwedge_{\mathcal{I}} F}{\Gamma \vdash^{\downarrow} \pi_j\ F}\ \textsc{AndInf-E}$$

$$\frac{\Gamma \vdash^{\uparrow} \pi_j\ F}{\Gamma \vdash^{\uparrow} \bigvee_{\mathcal{I}} F}\ \textsc{OrInf-I} \qquad \frac{\Gamma \vdash^{\downarrow} \bigvee_{\mathcal{I}} F \qquad \Gamma, \pi_i\ F \vdash^{\uparrow} C \text{ for every } i \in \mathcal{I}}{\Gamma \vdash^{\uparrow} C}\ \textsc{OrInf-E}$$

$$\frac{\Gamma \vdash^{\uparrow} P\ x \qquad x \notin \mathsf{fv}(\Gamma, P)}{\Gamma \vdash^{\uparrow} \forall\ P}\ \textsc{All-I} \qquad \frac{\Gamma \vdash^{\downarrow} \forall\ P}{\Gamma \vdash^{\downarrow} P\ t}\ \textsc{All-E}$$

$$\frac{\Gamma \vdash^{\uparrow} P\ t}{\Gamma \vdash^{\uparrow} \exists\ P}\ \textsc{Ex-I} \qquad \frac{\Gamma \vdash^{\downarrow} \exists\ P \qquad \Gamma, P\ x \vdash^{\uparrow} A \qquad x \notin \mathsf{fv}(\Gamma, P, A)}{\Gamma \vdash^{\uparrow} A}\ \textsc{Ex-E}$$

$$\frac{\Gamma \vdash^{\downarrow} \bot}{\Gamma \vdash^{\uparrow} A}\ \textsc{Bot-E} \qquad \frac{}{\Gamma \vdash^{\downarrow} A \vee \neg A}\ \textsc{PEM}$$

Figure 1 Cut-free natural deduction rules of higher-order infinitary logic.

has a finite height. The formula $A \wedge B$ (respectively $A \vee B$) is defined by $\bigwedge_{\mathcal{I}} \{i \mapsto A_i\}_{\{0,1\}}$ (respectively $\bigvee_{\mathcal{I}} \{i \mapsto A_i\}_{\{0,1\}}$), where $A_0 = A$ and $A_1 = B$. The inference rules for the finite conjunction (And-I, And-EL, And-ER) and disjunction (Or-IL, Or-IR, Or-E) are particular cases of those for the infinite conjunction and disjunction.

We consider either classical logic CL or intuitionistic logic IL – in which we do not have the PEM rule. We write $\vdash_{\mathsf{L}}$ when we consider the inference rules of $\mathsf{L} \in \{\mathsf{CL}, \mathsf{IL}\}$.

The weakening, contraction and exchange inference rules are admissible for $\vdash^{\downarrow}$, $\vdash^{\uparrow}$ and $\vdash$.

▶ **Lemma 15** (Axiom Replacement). *The following rule is admissible:*

$$\frac{\Gamma \vdash^{\downarrow} A \qquad \Delta, A \vdash^{*} B}{\Delta, \Gamma \vdash^{*} B}\ \textit{Ax-R}$$

Proof. By induction on the proof of $\Delta, A \vdash^{*} B$. ◀

▶ **Example 16** (Natural numbers). Let us define $\mathsf{0}$ of type ι and the successor function s of type $\iota \to \iota$. We also set an infix equality symbol $=$ of type $\iota \to \iota \to o$, that satisfies the reflexivity axiom $\forall x.\ x = x$ and the Leibniz rule $\forall x. \forall y. \forall P.\ (x = y \wedge P\ x) \Rightarrow P\ y$. We encode the induction principle NatInd

$$\forall P.\ ((P\ \mathsf{0} \wedge \forall x.\ (P\ x \Rightarrow P\ (\mathsf{s}\ x))) \Rightarrow \forall x.\ P\ x)$$

using a higher-order quantification on any predicate P. We also encode an infinitary formula ItNat stating that natural numbers are iterated successors of zero

$$\forall x.\ \bigvee\nolimits_{\mathbb{N}} \{n \mapsto x = \mathsf{s}^n\ \mathsf{0}\}_{\mathbb{N}}$$

where $\mathsf{s}^n\ \mathsf{0}$ means that the successor s has been applied n times to $\mathsf{0}$. In higher-order infinitary logic, these two principles are equivalent.

- To prove that NatInd implies ItNat, we use the induction principle NatInd with the predicate $\lambda x.\ \bigvee_{\mathbb{N}} \{n \mapsto x = \mathsf{s}^n\ 0\}_{\mathbb{N}}$. We prove the base case with OrInf-I and $n = 0$. We prove the inductive step with OrInf-E, as once we assume that $x = \mathsf{s}^n\ 0$ for some $n \in \mathbb{N}$, it is easy to show that $\mathsf{s}\ x = \mathsf{s}^{n+1}\ 0$, and thus $\bigvee_{\mathbb{N}} \{n \mapsto \mathsf{s}\ x = \mathsf{s}^n\ 0\}_{\mathbb{N}}$.
- To prove that ItNat implies NatInd, we use OrInf-E with the infinite disjunction ItNat. For every $n \in \mathbb{N}$, we have to prove $P\ x$ assuming $(P\ 0 \wedge \forall x.\ (P\ x \Rightarrow P\ (\mathsf{s}\ x))$ and $x = \mathsf{s}^n\ 0$. Using the Leibniz principle, it remains to show $P\ (\mathsf{s}^n\ 0)$. To do so, we apply n times the induction step from $P\ 0$.

We can therefore conveniently use either of the two principles.

Cuts. Unlike in sequent calculus, cuts in natural deduction may be defined with various degrees of looseness. The restricted notion of cut is that of an introduction rule immediately followed by its corresponding elimination rule. The rules of Figure 1 precludes these cuts. For instance, the following proof, that contains a cut, cannot be expressed because the conclusion of And-I is $A, B \vdash^{\uparrow} A \wedge B$ while the premise of And-EL is $A, B \vdash^{\downarrow} A \wedge B$.

$$\dfrac{\dfrac{\dfrac{}{A, B \vdash A}\ \textsc{Ax} \quad \dfrac{}{A, B \vdash B}\ \textsc{Ax}}{A, B \vdash A \wedge B}\ \textsc{And-I}}{A, B \vdash A}\ \textsc{And-EL}$$

The rules of Figure 1 additionally precludes the so-called *commutative cuts*, that separate the introduction and elimination rules on a single connective with a series of Or-E, OrInf-E and Ex-E rules. This is because we have restricted the conclusion sequent of those three rules, plus the Bot-E rule, to be cut-free ($\vdash^{\uparrow}$), rather than letting it being neutral ($\vdash^{\downarrow}$), as it would be the case for lighter cut-elimination processes. For instance, the following proof with $\Gamma = A, B, C \vee D$, has a commutative cut.

$$\dfrac{\dfrac{}{\Gamma \vdash C \vee D}\ \textsc{Ax} \quad \dfrac{\dfrac{\dfrac{}{\Gamma, C \vdash A}\ \textsc{Ax} \quad \dfrac{}{\Gamma, C \vdash B}\ \textsc{Ax}}{\Gamma, C \vdash A \wedge B}\ \textsc{And-I} \quad \dfrac{\vdots}{\Gamma, D \vdash A \wedge B}}{\Gamma \vdash A \wedge B}\ \textsc{Or-E}}{\Gamma \vdash A}\ \textsc{And-EL}$$

It cannot be expressed with the cut-free rules, because the conclusion of Or-E is $\Gamma \vdash^{\uparrow} A \wedge B$ while the premise of And-EL is $\Gamma \vdash^{\downarrow} A \wedge B$.

▶ Remark 17. The cut-elimination result for natural deduction can be readily transferred to sequent calculus, using a translation from cut-free natural deduction to cut-free sequent calculus [8].

4 Soundness, Strong Completeness and Cut Elimination

In this section, we introduce a sound notion of model based on Heyting algebras, and we prove a strong version of completeness that entails the cut-elimination theorem for higher-order infinitary logic in natural deduction. This proof relies on building a specific model, in which veracity essentially amounts to provability. As we must account for impredicativity and intensionality, we wrap this idea with refinement of the V-complexes technique [29, 35, 1]. We work in the same spirit as [5, 16, 17].

4.1 Models for Higher-Order Infinitary Logic

We recall that a *Heyting algebra* is an ordered lattice $(\mathcal{H}, \leq, \wedge, \vee)$, where $\leq$ is a partial order on $\mathcal{H}$, $\wedge$ is a binary greatest lower bound operator, called meet, and $\vee$ is a binary largest upper bound operator, called join. In addition, $\mathcal{H}$ is equipped with a relative pseudo-complement operator $\to$, that verifies, for any $a, b, c \in \mathcal{H}$, the *implication property*: $a \wedge b \leq c$ if, and only if $a \leq b \to c$. A *complete Heyting algebra* is a Heyting algebra whose base lattice is complete, that is to say, that enjoys arbitrary meets ($\bigwedge$) and joins ($\bigvee$). The global maximum (empty meet) and minimum (empty joint) will be denoted $\top_{\mathcal{H}}$ and $\bot_{\mathcal{H}}$, respectively. The support $\mathcal{H}$ of the Heyting algebra will also denote the algebra itself. In any Heyting algebra, $\wedge$ and $\vee$ distribute over each other. Moreover, $\wedge$ distributes over $\bigvee$, whenever the latter exists.

To establish models of higher-order infinitary logic, we have to take into account its *non-extensional* nature. To illustrate this feature, let P be a predicate of type $o \to o$. Without any further information on P, it is impossible to show $P(\bot) \vdash P(\bot \wedge \bot)$, although $\bot$ and $\bot \wedge \bot$ are logically equivalent. Therefore, the term $\bot$ must receive a different interpretation than $\bot \wedge \bot$, although semantically we have $\bot_{\mathcal{H}} = \bot_{\mathcal{H}} \wedge_{\mathcal{H}} \bot_{\mathcal{H}}$ in any Heyting algebra. This imposes a separation of the domain of interpretation of formulas M_o from the Heyting algebra $\mathcal{H}$. The latter contains truth values, while the former carries more information. Both are linked through a truth-value extraction function $\omega : \mathsf{M}_o \to \mathcal{H}$. To do so, we resort to Heyting applicative structures and adapt the notion of model introduced in [5, 16, 17].

▶ **Definition 18** (Heyting Applicative Structure). *A Heyting applicative structure is a quintuple* $\langle \mathsf{M}, \mathsf{App}, \mathsf{Const}, \omega, \mathcal{H} \rangle$ *composed of:*

- *a set* M_τ *for each type* τ*, called domain,*
- *a function* $\mathsf{App}_{\tau,\sigma} : \mathsf{M}_{\tau\to\sigma} \times \mathsf{M}_\tau \to \mathsf{M}_\sigma$ *for each pair of types* (τ, σ)*, and a function* $\mathsf{App}_{o_\mathcal{I}} : \mathsf{M}_{o_\mathcal{I}} \times \mathcal{I} \to \mathsf{M}_o$
- *an interpretation function* Const_τ *for each type* τ*, taking constants of type* τ *into* M_τ*,*
- *a complete Heyting algebra* $\mathcal{H}$*,*
- *a function* $\omega : \mathsf{M}_o \to \mathcal{H}$*,*

such that the following conditions are satisfied, for every $m \in \mathsf{M}_{\tau\to o}$ *and every* $m' \in \mathsf{M}_{o_\mathcal{I}}$*:*

$$
\begin{aligned}
&\omega(\mathsf{Const}(\bot)) && = \bot_{\mathcal{H}} \\
&\omega(\mathsf{App}(\mathsf{App}(\mathsf{Const}(\Rightarrow), d_1), d_2)) && = \omega(d_1) \to_{\mathcal{H}} \omega(d_2) \\
&\omega(\mathsf{App}(\mathsf{Const}(\exists_\tau), m)) && = \textstyle\bigvee_{d \in \mathsf{M}_\tau} \omega(\mathsf{App}(m, d)) \\
&\omega(\mathsf{App}(\mathsf{Const}(\forall_\tau), m)) && = \textstyle\bigwedge_{d \in \mathsf{M}_\tau} \omega(\mathsf{App}(m, d)) \\
&\omega(\mathsf{App}(\mathsf{Const}(\textstyle\bigwedge_\mathcal{I}), m')) && = \textstyle\bigwedge_{i \in \mathcal{I}} \omega(\mathsf{App}(m', i)) \\
&\omega(\mathsf{App}(\mathsf{Const}(\textstyle\bigvee_\mathcal{I}), m')) && = \textstyle\bigvee_{i \in \mathcal{I}} \omega(\mathsf{App}(m', i))
\end{aligned}
$$

When interpreting formulas in a model, and as soon as we have complete Heyting algebras, there is no much difference between quantifiers and the infinitary connectives. In both cases, we use arbitrary meets and joins: indexed by $d \in \mathsf{M}_\tau$ for the quantifiers and indexed by $i \in \mathcal{I}$ for the infinitary connectives.

An assignment is a function φ with finite support, such that for any variable x of type τ of the support, $\varphi(x)$ is an element of M_τ. The assignment $\varphi[d/x]$ is identical to φ, save on x, where $\varphi[d/x](x) = d$.

▶ **Definition 19** (Model). *Given a Heyting applicative structure, a model is a function* $[\![_]\!]_$ *that maps a term of type* τ *and an assignment to* M_τ*, such that:*

$$
\begin{array}{rcl}
[\![c]\!]_\varphi & = & \mathsf{Const}(c) \\
[\![x]\!]_\varphi & = & \varphi(x) \\
[\![t\ u]\!]_\varphi & = & \mathsf{App}([\![t]\!]_\varphi, [\![u]\!]_\varphi) \\
[\![\pi_j\ F]\!]_\varphi & = & \mathsf{App}([\![F]\!]_\varphi, j) \\
[\![\lambda x_\tau.\ t_\sigma]\!]_\varphi & & \textit{is the unique member of } \mathsf{M}_{\tau\to\sigma} \textit{ such that} \\
 & & \mathsf{App}([\![\lambda x_\tau.\ t_\sigma]\!]_\varphi, d) = [\![t]\!]_{\varphi[d/x]} \textit{ for every } d \in \mathsf{M}_\tau \\
[\![\{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi & & \textit{is the unique member of } \mathsf{M}_{o_\mathcal{I}} \textit{ such that} \\
 & & \mathsf{App}([\![\{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi, j) = [\![A_j]\!]_\varphi \textit{ for every } j \in \mathcal{I} \\
[\![t]\!]_\varphi & = & [\![u]\!]_\varphi \qquad\qquad \textit{if } t \equiv u
\end{array}
$$

By model, we will often refer to the combination of the Heyting applicative structure, the model interpretation function above, and the assignment. With this notion, we can state the soundness theorem. When $\Gamma = A_1, \ldots, A_n$, we write $\bigwedge \Gamma$ for the formula $A_1 \wedge \ldots \wedge A_n$.

▶ **Theorem 20** (Soundness). *If* $\Gamma \vdash A$ *then for any model* $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \omega([\![A]\!]_\varphi)$.

Proof. By induction on the derivation. We show the most interesting cases:

- AndInf-I: By induction, we have $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \omega([\![\pi_i\ F]\!]_\varphi)$ for every $i \in \mathcal{I}$. As we are in a complete Heyting algebra, we have $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \bigwedge_{i\in\mathcal{I}} \omega([\![\pi_i\ F]\!]_\varphi)$, that is $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \omega([\![\bigwedge_\mathcal{I} F]\!]_\varphi)$.
- OrInf-E: By induction, $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \bigvee_{i\in\mathcal{I}} \omega([\![\pi_i\ F]\!]_\varphi)$ and $\omega([\![\bigwedge \Gamma]\!]_\varphi) \wedge \omega([\![\pi_i\ F]\!]_\varphi) \leq \omega([\![C]\!]_\varphi)$ for any $i \in \mathcal{I}$. We derive $\omega([\![\pi_i\ F]\!]_\varphi) \leq \omega([\![\bigwedge \Gamma]\!]_\varphi) \to \omega([\![C]\!]_\varphi)$ for any $i \in \mathcal{I}$, and therefore $\bigvee_{i\in\mathcal{I}} \omega([\![\pi_i\ F]\!]_\varphi) \leq \omega([\![\bigwedge \Gamma]\!]_\varphi) \to \omega([\![C]\!]_\varphi)$. We get $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \omega([\![\bigwedge \Gamma]\!]_\varphi) \to \omega([\![C]\!]_\varphi)$, and thus $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \omega([\![C]\!]_\varphi)$. ◀

▶ Remark 21. Note that this theorem is seemingly stated and proved for intuitionistic logic and Heyting algebras, but it also applies to classical logic, provided that we restrict the Heyting algebras $\mathcal{H}$ of the models to be Boolean algebras. More generally, all the results and construction of this section apply to both the intuitionistic and classical cases. For instance, all the algebras that we construct can easily be shown to be Boolean provided that provability itself is classical.

Instead of showing completeness for the natural deduction with cuts thanks to the Lindenbaum algebra, we will construct another Heyting algebra, that allows to show a stronger form of the completeness theorem for the cut-free natural deduction. We define the cut-free-provability algebra in Section 4.2, and the cut-free-provability model in Section 4.3.

4.2 The Cut-Free-Provability Heyting Algebra

We build a cut-free Heyting algebra following [16, 17]. It is based on Schütte's semi-valuations [31, 32], that have been first used to show cut admissibility for higher-order classical logic [35, 29, 1]. Semi-valuations have been subsequently extended to higher-order intuitionistic logic [24, 5] and are nowadays a versatile tool, that can be fine-tuned a number of ways and fits many logics [16, 17, 24, 27]. In this paper, semi-valuations are embodied by the upper and lower values $\lceil_\rceil$ and $\lfloor_\rfloor$, that respectively represent the maximal and minimal constraints that our target model interpretation should respect. This is the essence of the model construction of Definition 28, that relies itself on Lemma 27.

▶ **Definition 22** (Upper Value). *The upper value of* $\Gamma \vdash A$*, denoted* $\lceil \Gamma \vdash A \rceil$*, is the set of contexts* $\{\Delta \mid \Gamma, \Delta \vdash^\uparrow A \textit{ is derivable}\}$*. We write* $\lceil A \rceil$ *for* $\lceil \vdash A \rceil$*.*

▶ **Definition 23** (Cut-Free-Provability Algebra). *Let $\mathcal{H}$ be the least set containing the $\lceil \Gamma \vdash A \rceil$ for any Γ, A, that is closed by arbitrary intersection. Let $\leq$ be $\subseteq$ and $\wedge$ be $\cap$. Let arbitrary meets and joins of any set of elements $S \subseteq \mathcal{H}$ be:*

- $\bigwedge S = \bigcap S$, *the intersection of the elements of* S,
- $\bigvee S = \bigcap\{c \in \mathcal{H} \mid \forall s \in S, s \leq c\}$, *the pseudo-union.*

For any $a, b \in \mathcal{H}$, we let $a \to b = \bigvee\{c \in \mathcal{H} \mid a \wedge c \leq b\}$.

Note that for any $S \subseteq \mathcal{H}$, the set $\{c \in \mathcal{H} \mid \forall s \in S, s \leq c\}$ is not empty, as it contains at least the set of all contexts. So arbitrary meets and joins always exist. Moreover, the definition of arbitrary joins can be slightly simplified and proved to be equal to $\bigvee S = \bigcap\{\lceil \Gamma \vdash A \rceil \mid \forall s \in S, s \leq \lceil \Gamma \vdash A \rceil\}$. This is because any $c \in \mathcal{H}$ is of the form $\bigcap_{k \in K} \lceil \Gamma_k \vdash A_k \rceil$. In the sequel, we will use either of those characterizations.

▶ **Lemma 24.** *$\mathcal{H}$ is a complete Heyting algebra.*

Proof. By construction, $\mathcal{H}$ is an ordered lattice that enjoys arbitrary joins and meets. We must check the implication property: $a \wedge b \leq c$ if and only if $a \leq b \to c$. The direct implication follows from basic lattice calculation. Let us show the reverse implication. Let $\Gamma \in a \wedge b$ and suppose that $a \leq b \to c$. By definition of $\mathcal{H}$, we have $c = \bigcap_{k \in K} \lceil \Sigma_k \vdash C_k \rceil$. We want to show that $\Sigma_k, \Gamma \vdash^{\uparrow} C_k$ for each $k \in K$. We have $\Gamma \in a \leq b \to c = \bigvee\{d \in \mathcal{H} \mid b \wedge d \leq c\}$.

As an intermediate step, let us prove that $d \leq \lceil \Sigma_k, \Gamma \vdash C_k \rceil$ for any d such that $b \wedge d \leq c$. Let $\Delta \in d$. Upper values are provability-based, and therefore elements of $\mathcal{H}$ admit context weakening. From $\Gamma \in b$ and $\Delta \in d$, we thus get $\Gamma, \Delta \in b \wedge d$. This entails $\Gamma, \Delta \in c$, i.e. $\Delta \in \lceil \Sigma_k, \Gamma \vdash C_k \rceil$. Thus $d \leq \lceil \Sigma_k, \Gamma \vdash C_k \rceil$ for any d such that $b \wedge d \leq c$.

Since $\Gamma \in \bigvee\{d \in \mathcal{H} \mid b \wedge d \leq c\}$, we derive $\Gamma \in \lceil \Sigma_k, \Gamma \vdash C_k \rceil$ and therefore $\Sigma_k, \Gamma, \Gamma \vdash^{\uparrow} C_k$. By contraction, we get $\Sigma_k, \Gamma \vdash^{\uparrow} C_k$. ◀

▶ **Definition 25** (Lower Value). *Let $s \in \mathcal{H}$ be a set of contexts. Its lower value is the least element of $\mathcal{H}$ that contains s, i.e., $\lfloor s \rfloor = \bigcap\{c \in \mathcal{H} \mid s \subseteq c\}$. We write $\lfloor A \rfloor$ for $\lfloor \{\{A\}\} \rfloor$.*

Lower values can be shown to be equal to $\lfloor s \rfloor = \bigcap\{\lceil \Gamma \vdash A \rceil \in \mathcal{H} \mid s \subseteq \lceil \Gamma \vdash A \rceil\}$. The rationale behind $\lfloor A \rfloor$ is the following: $\Gamma \in \lfloor A \rfloor$ if and only if $\Delta, \Gamma \vdash^{\uparrow} B$ as soon as $\Delta, A \vdash^{\uparrow} B$. Lower values have strong links with the characterization of neutral proofs.

▶ **Lemma 26.** *If $\Gamma \vdash^{\downarrow} A$, then $\Gamma \in \lfloor A \rfloor$.*

Proof. Direct consequence of Lemma 15. ◀

The following property characterizes the lower and upper values for the connectives and quantifiers. It will be of crucial importance, as identified by [27, 5]. Note that the inequalities for implication mix lower and upper values. More generally, for any formula A, we have the inequality $\lfloor A \rfloor \leq \lceil A \rceil$.

▶ **Lemma 27** (Main Lemma). *The lower and upper values respect the following constraints:*

$$
\begin{array}{rcl@{\qquad\qquad}rcl}
\lfloor \bot \rfloor & = & \bot_{\mathcal{H}} & & & \\
\lfloor A \Rightarrow B \rfloor & \leq & \lceil A \rceil \to \lfloor B \rfloor & \lfloor A \rfloor \to \lceil B \rceil & \leq & \lceil A \Rightarrow B \rceil \\
\lfloor \forall_\tau\, P \rfloor & \leq & \bigwedge_{t \in \Lambda_\tau} \lfloor P\ t \rfloor & \bigwedge_{t \in \Lambda_\tau} \lfloor P\ t \rfloor & \leq & \lceil \forall_\tau\, P \rceil \\
\lfloor \exists_\tau\, P \rfloor & \leq & \bigvee_{t \in \Lambda_\tau} \lfloor P\ t \rfloor & \bigvee_{t \in \Lambda_\tau} \lceil P\ t \rceil & \leq & \lceil \exists_\tau\, P \rceil \\
\lfloor \bigwedge_{\mathcal{I}} F \rfloor & \leq & \bigwedge_{i \in \mathcal{I}} \lfloor \pi_i\ F \rfloor & \bigwedge_{i \in \mathcal{I}} \lceil \pi_i\ F \rceil & \leq & \lceil \bigwedge_{\mathcal{I}} F \rceil \\
\lfloor \bigvee_{\mathcal{I}} F \rfloor & \leq & \bigvee_{i \in \mathcal{I}} \lfloor \pi_i\ F \rfloor & \bigvee_{i \in \mathcal{I}} \lceil \pi_i\ F \rceil & \leq & \lceil \bigvee_{\mathcal{I}} F \rceil
\end{array}
$$

Proof. Each case involves the corresponding cut-free natural deduction rule. We only show the cases for infinite disjunction and conjunction, as they are the added value of our paper, compared to the previous works [16]. Moreover, those four cases also faithfully represent the cases for the quantifiers.

- Infinite conjunction (lower): Let $\Gamma \in \lfloor \bigwedge_{\mathcal{I}} F \rfloor$ and $j \in \mathcal{I}$. Suppose $\Delta, \pi_j\ F \vdash^{\uparrow} B$. Combining the proof
$$\dfrac{\dfrac{\dfrac{}{\bigwedge_{\mathcal{I}} F \vdash^{\downarrow} \bigwedge_{\mathcal{I}} F}\ \textsc{Ax}}{\bigwedge_{\mathcal{I}} F \vdash^{\downarrow} \pi_j\ F}\ \textsc{AndInf-E} \qquad \Delta, \pi_j\ F \vdash^{\uparrow} B}{\Delta, \bigwedge_{\mathcal{I}} F \vdash^{\uparrow} B}\ \textsc{Ax-R}$$
and the defining property of $\Gamma \in \lfloor \bigwedge_{\mathcal{I}} F \rfloor$, we get $\Delta, \Gamma \vdash^{\uparrow} B$. Therefore $\Gamma \in \lfloor \pi_j\ F \rfloor$ for every $j \in \mathcal{I}$. It follows that $\lfloor \bigwedge_{\mathcal{I}} F \rfloor \leq \bigwedge_{i \in \mathcal{I}} \lfloor \pi_i\ F \rfloor$.
- Infinite conjunction (upper): Let $\Gamma \in \bigwedge_{i \in \mathcal{I}} \lceil \pi_i\ F \rceil$. Using the proof
$$\dfrac{\Gamma \vdash^{\uparrow} \pi_i\ F \text{ for every } i \in \mathcal{I}}{\Gamma \vdash^{\uparrow} \bigwedge_{\mathcal{I}} F}\ \textsc{AndInf-I}$$
we get $\Gamma \in \lceil \bigwedge_{\mathcal{I}} F \rceil$.
- Infinite disjunction (lower): Let $\Gamma \in \lfloor \bigvee_{\mathcal{I}} F \rfloor$. Let $\bigcap_{k \in K} \lceil \Delta_k \vdash B_k \rceil$ be an upper bound of all the $\lfloor \pi_i\ F \rfloor$. Combining the following proof
$$\dfrac{\dfrac{}{\Delta_k, \bigvee_{\mathcal{I}} F \vdash^{\downarrow} \bigvee_{\mathcal{I}} F}\ \textsc{Ax} \qquad \dfrac{\Delta_k, \pi_i\ F \vdash^{\uparrow} B_k \text{ for every } i \in \mathcal{I}}{\Delta_k, \bigvee_{\mathcal{I}} F, \pi_i\ F \vdash^{\uparrow} B_k \text{ for every } i \in \mathcal{I}}\ \text{Weakening}}{\Delta_k, \bigvee_{\mathcal{I}} F \vdash^{\uparrow} B_k}\ \textsc{OrInf-E}$$
and $\Gamma \in \lfloor \bigvee_{\mathcal{I}} F \rfloor$, we get $\Delta_k, \Gamma \vdash^{\uparrow} B_k$. It follows that $\Gamma \in \bigcap_{k \in K} \lceil \Delta_k \vdash B_k \rceil$, and therefore $\lfloor \bigvee_{\mathcal{I}} F \rfloor \leq \bigcap_{k \in K} \lceil \Delta_k \vdash B_k \rceil$. As $\bigvee_{i \in \mathcal{I}} \lfloor \pi_i\ F \rfloor$ is an upper bound of all the $\lfloor \pi_i\ F \rfloor$, we get $\lfloor \bigvee_{\mathcal{I}} F \rfloor \leq \bigvee_{i \in \mathcal{I}} \lfloor \pi_i\ F \rfloor$.
- Infinite disjunction (upper): Let $j \in \mathcal{I}$ and $\Gamma \in \lceil \pi_j\ F \rceil$. Using the proof
$$\dfrac{\Gamma \vdash^{\uparrow} \pi_j\ F}{\Gamma \vdash^{\uparrow} \bigvee_{\mathcal{I}} F}\ \textsc{OrInf-I}$$
we get $\Gamma \in \lceil \bigvee_{\mathcal{I}} F \rceil$. Therefore $\lceil \pi_j\ F \rceil \leq \lceil \bigvee_{\mathcal{I}} F \rceil$ for every $j \in \mathcal{I}$. It follows that $\bigvee_{i \in \mathcal{I}} \lceil \pi_i\ F \rceil \leq \lceil \bigvee_{\mathcal{I}} F \rceil$. ◀

4.3 The Cut-Free-Provability Model

To build the cut-free-provability model, we first define the domains M_τ. As our logic is intensional, it is important to have $\mathsf{M}_o \neq \mathcal{H}$. This calls for the so-called V-complex construction, defined by Takahashi and Prawitz to show cut admissibility of classical higher-order logic [29, 35, 1]. The core of this construction is an entanglement of a syntactic and a semantic value.

▶ **Definition 28** (V-complexes). *We define the domain M_τ by induction on τ, as sets of pairs of the form $\langle [t], \nu \rangle$, where t is a term of type τ and $[t]$ is its $\equiv$-equivalence class, and the following conditions hold on ν:*

- *for type ι, ν is a fixed constant, traditionally named ι,*
- *for type o, $\nu \in \mathcal{H}$ and $\lfloor t \rfloor \leq \nu \leq \lceil t \rceil$,*
- *for type $\tau \to \sigma$, ν is a function of type $\mathsf{M}_\tau \to \mathsf{M}_\sigma$ such that, for any $\langle [u], \nu_u \rangle \in \mathsf{M}_\tau$, there exists some ν' such that $\nu(\langle [u], \nu_u \rangle) = \langle [t\ u], \nu' \rangle \in \mathsf{M}_\sigma$,*
- *for type $o_{\mathcal{I}}$, ν is a function of type $\mathcal{I} \to \mathsf{M}_o$ such that, for any $i \in \mathcal{I}$, there exists some ν_i such that $\nu(i) = \langle [\pi_i\ t], \nu_i \rangle \in \mathsf{M}_o$.*

We need to ensure that the domains M_τ are well-formed. We first check that the definition does not depend on the chosen representatives of the $\equiv$-equivalence classes:

- if $[t] = [t']$ then $\lfloor t \rfloor = \lfloor t' \rfloor$ and $\lceil t \rceil = \lceil t' \rceil$ (because of the Conv rule),
- if $t' \in [t]$ and $u' \in [u]$ then $[t'\ u'] = [t\ u]$,
- if $t' \in [t]$ then $[\pi_i\ t'] = [\pi_i\ t]$ for every $i \in \mathcal{I}$.

We also check the non-emptiness of these domains, thanks to the following lemma.

▶ **Lemma 29** (Canonical Representatives). *For any term t of type τ, there exists $\rho(t)$ such that $\langle [t], \rho(t)\rangle \in \mathsf{M}_\tau$.*

Proof. The definition of $\rho(_)$ is done by induction over the type τ. The verification that it meets the requirements is performed altogether and is immediate, given that all the manipulated structures are invariant under $\equiv$. We let $\rho(t_\iota) = \iota$, $\rho(t_o) = \lceil t_o \rceil$[1], $\rho(t_{\tau\to\sigma})$ be the function $\langle [u_\tau], \nu_u\rangle \mapsto \langle [t\ u], \rho(t\ u)\rangle$, and $\rho(t_{o_\mathcal{I}})$ be the function $i \mapsto \langle [\pi_i\ t], \rho(\pi_i\ t)\rangle$. ◀

We now define the interpretation of constants. We write $\|_\|_1$ (respectively $\|_\|_2$) for the projection of a member of M_τ on its first (respectively second) component. Following the definition of the domains, $\|_\|_1$ returns an equivalence class. Since all our constructions are invariant under $\equiv$, it is sound, for any $d \in \mathsf{M}_\tau$, to define the term $\|d\|_r$ as *a* representative of the equivalence class $\|d\|_1$.

▶ **Definition 30** (Interpretation of Constants). *We let:*

$$
\begin{aligned}
\mathsf{Const}(\bot) &= \langle [\bot], \bot_\mathcal{H}\rangle \\
\mathsf{Const}(\Rightarrow) &= \langle [\Rightarrow], \boldsymbol{\lambda}\langle [A], a\rangle.\langle [\Rightarrow A], \boldsymbol{\lambda}\langle [B], b\rangle.\langle [\Rightarrow A\ B], a \to b\rangle\rangle\rangle \\
\mathsf{Const}(\forall_\tau) &= \langle [\forall_\tau], \boldsymbol{\lambda}\langle [P], p\rangle.\langle [\forall\, P], \textstyle\bigwedge_{d\in\mathsf{M}_\tau} \|p(d)\|_2\rangle\rangle \\
\mathsf{Const}(\exists_\tau) &= \langle [\exists_\tau], \boldsymbol{\lambda}\langle [P], p\rangle.\langle [\exists\, P], \textstyle\bigvee_{d\in\mathsf{M}_\tau} \|p(d)\|_2\rangle\rangle \\
\mathsf{Const}(\textstyle\bigwedge_\mathcal{I}) &= \langle [\textstyle\bigwedge_\mathcal{I}], \boldsymbol{\lambda}\langle [F], f\rangle.\langle [\textstyle\bigwedge_\mathcal{I} F], \bigwedge_{i\in\mathcal{I}} \|f(i)\|_2\rangle\rangle \\
\mathsf{Const}(\textstyle\bigvee_\mathcal{I}) &= \langle [\textstyle\bigvee_\mathcal{I}], \boldsymbol{\lambda}\langle [F], f\rangle.\langle [\textstyle\bigvee_\mathcal{I} F], \bigvee_{i\in\mathcal{I}} \|f(i)\|_2\rangle\rangle \\
\mathsf{Const}(c) &= \langle [c], \rho(c)\rangle \textit{ for non-logical constants}
\end{aligned}
$$

where $\boldsymbol{\lambda}$ denotes the function formation primitive of type $\mathsf{M}_\tau \to \mathsf{M}_\sigma$.

▶ **Lemma 31.** *The interpretation of constants are members of their respective domains.*

Proof. For non-logical constants, the claim holds by Lemma 29. For logical constants, proving the claim boils down to checking that the end truth value respects the conditions of being a member of M_o, that imposes the truth value to be bounded by the lower and upper values. For this, we use Lemma 27.

- Contradiction: We directly have $\lfloor \bot \rfloor = \bot_\mathcal{H}$.
- Implication: By definition, $\lfloor A \rfloor \le a \le \lceil A \rceil$ and $\lfloor B \rfloor \le b \le \lceil B \rceil$. Remember that $\to$ is contravariant in its left argument (if $c \le c'$, then $c' \to d \le c \to d$). Therefore we have:

$$\lfloor A \Rightarrow B \rfloor \le \lceil A \rceil \to \lfloor B \rfloor \le a \to b \le \lfloor A \rfloor \to \lceil B \rceil \le \lceil A \to B \rceil$$

- Universal quantifier: By definition of $\langle [P], p\rangle$, we have $p(d) \in \mathsf{M}_o$ for any $d \in \mathsf{M}_\tau$, that is to say $\lfloor P\ \|d\|_r \rfloor \le \|p(d)\|_2 \le \lceil P\ \|d\|_r \rceil$. By Lemma 29, all $t \in \Lambda_\tau$ are represented in M_τ by at least $\langle [t], \rho(t)\rangle$, and lower and upper values are stable by $\equiv$. Thus:

$$
\begin{aligned}
\lfloor \forall_\tau\ P \rfloor \le \textstyle\bigwedge_{t\in\Lambda_\tau} \lfloor P\ t \rfloor = \bigwedge_{d\in\mathsf{M}_\tau} \lfloor P\ \|d\|_r \rfloor &\le \textstyle\bigwedge_{d\in\mathsf{M}_\tau} \|p(d)\|_2 \\
&\le \textstyle\bigwedge_{d\in\mathsf{M}_\tau} \lceil P\ \|d\|_r \rceil = \bigwedge_{t\in\Lambda_\tau} \lceil P\ t \rceil \le \lceil \forall_\tau\ P \rceil
\end{aligned}
$$

[1] The choice of $\lceil _ \rceil$ is arbitrary and represents, in certain circumstances, a valuable degree of freedom.

- Infinite disjunction: By definition of $\langle [F], f\rangle$, we have $\lfloor \pi_i\ F\rfloor \leq \|f(i)\|_2 \leq \lceil \pi_i\ F\rceil$ for any $i \in \mathcal{I}$. Therefore:

$$\lfloor \bigvee_{\mathcal{I}} F\rfloor \leq \bigvee_{i\in\mathcal{I}} \lfloor \pi_i\ F\rfloor \leq \bigvee_{i\in\mathcal{I}} \|f(i)\|_2 \leq \bigvee_{i\in\mathcal{I}} \lceil \pi_i\ F\rceil \leq \lceil \bigvee_{\mathcal{I}} F\rceil$$

- The cases for the existential quantifier and the infinite disjunction are similar. ◀

We build the Heyting applicative structure by defining ω as the second projection $\|_\|_2$, and App so that $\mathsf{App}(m, d) = \|m\|_2(d)$ and $\mathsf{App}(m', i) = \|m'\|_2(i)$.

▶ **Lemma 32.** *Let ω be $\langle [_], \nu\rangle \mapsto \nu$, and App be $(\langle [_], \nu\rangle\ , d) \mapsto \nu(d)$ and $(\langle [_], \nu\rangle\ , i) \mapsto \nu(i)$. Then $\langle \mathsf{M}, \mathsf{App}, \mathsf{Const}, \omega, \mathcal{H}\rangle$ is a Heyting applicative structure.*

Proof. The conditions for being a Heyting applicative structure are immediate using the definitions of Const, App and ω. We only highlight a few cases.

$$\begin{aligned}\omega(\mathsf{App}(\mathsf{App}(\mathsf{Const}(\Rightarrow), d_1), d_2)) &= \omega(\langle [\Rightarrow \|d_1\|_r\ \|d_2\|_r], \|d_1\|_2 \to \|d_2\|_2\rangle)\\ &= \omega(d_1) \to \omega(d_2)\end{aligned}$$

$$\begin{aligned}\omega(\mathsf{App}(\mathsf{Const}(\exists_\tau), m)) &= \omega(\langle [\exists_\tau \|m\|_r], \bigvee_{d\in\mathsf{M}_\tau} \|\|m\|_2(d)\|_2\rangle)\\ &= \bigvee_{d\in\mathsf{M}_\tau} \omega(\mathsf{App}(m, d))\end{aligned}$$

$$\begin{aligned}\omega(\mathsf{App}(\mathsf{Const}(\textstyle\bigwedge_{\mathcal{I}}), m')) &= \omega(\langle [\textstyle\bigwedge_{\mathcal{I}} \|m'\|_r], \bigwedge_{i\in\mathcal{I}} \|\|m'\|_2(i)\|_2\rangle)\\ &= \textstyle\bigwedge_{i\in I} \omega(\mathsf{App}(m', i))\end{aligned}$$ ◀

For any assignment φ, we let $\|\varphi\|_r$ be any substitution that maps the variables x of the support of φ to $\|\varphi(x)\|_r$. Note that $\|\varphi\|_r$ is not uniquely determined because each $\equiv$-equivalence class has many representatives.

▶ **Lemma 33.** *There exists a unique function $[\![_]\!]_$ that satisfies all the conditions of Definition 19, except the last one. It satisfies $\|[\![t]\!]_\varphi\|_1 = [t\|\varphi\|_r]$ for any term t with variables in the support of φ.*

Proof. We build $[\![_]\!]_$ by induction as the unique function that satisfies the conditions of Definition 19, except the last one, and we check $\|[\![t]\!]_\varphi\|_1 = [t\|\varphi\|_r]$ for each case. The cases for constants and variables are immediate. The case for applications follows from the induction hypotheses. The case for abstractions is similar to the case for families of propositions.

- Projections: By induction hypothesis, $\|[\![F]\!]_\varphi\|_1 = [F\|\varphi\|_r]$. By definition of $\mathsf{M}_{o_\mathcal{I}}$, it follows that $[\![F]\!]_\varphi = \langle [F\|\varphi\|_r], i \mapsto \langle [\pi_i\ F\|\varphi\|_r], \nu_i\rangle\rangle$, with $\langle [\pi_i\ F\|\varphi\|_r], \nu_i\rangle \in \mathsf{M}_o$. We set $[\![\pi_j\ F]\!]_\varphi := \mathsf{App}([\![F]\!]_\varphi, j) = \langle [\pi_j\ F\|\varphi\|_r], \nu_j\rangle$. Therefore we have $[\![\pi_j\ F]\!]_\varphi \in \mathsf{M}_o$ and $\|[\![\pi_j\ F]\!]_\varphi\|_1 = [\pi_j\ F\|\varphi\|_r] = [(\pi_j\ F)\|\varphi\|_r]$.
- Families: By definition of $\mathsf{M}_{o_\mathcal{I}}$, $[\![\{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi$ should be of the form $\langle [t], i \mapsto \langle [\pi_i\ t], \nu_i\rangle\rangle$, with $\langle [\pi_i\ t], \nu_i\rangle \in \mathsf{M}_o$. By induction hypothesis, every $[\![A_j]\!]_\varphi$ is uniquely defined and satisfies $\|[\![A_j]\!]_\varphi\|_1 = [A_j\|\varphi\|_r]$. For every $j \in \mathcal{I}$, we want $\mathsf{App}([\![\{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi, j) = [\![A_j]\!]_\varphi$, that is $\langle [\pi_j\ t], \nu_j\rangle = [\![A_j]\!]_\varphi$. This uniquely determines ν_j, ensures $\langle [\pi_j\ t], \nu_j\rangle \in \mathsf{M}_o$ and imposes $[\pi_j\ t] = \|[\![A_j]\!]_\varphi\|_1 = [A_j\|\varphi\|_r]$. Therefore $[t] = [\{i \mapsto \pi_i\ t\}_\mathcal{I}] = [\{i \mapsto A_i\|\varphi\|_r\}_\mathcal{I}]$, and we have $\|[\![\{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi\|_1 = [\{i \mapsto A_i\|\varphi\|_r\}_\mathcal{I}] = [\{i \mapsto A_i\}_\mathcal{I}\|\varphi\|_r]$. ◀

▶ Remark 34. In higher-order logic, the uniqueness of the function $[\![_]\!]_$ requires to consider the η-rule. In our extension of simple type theory, we additionally have to consider the rule $\{i \mapsto \pi_i\ F\}_\mathcal{I} \hookrightarrow F$.

▶ **Lemma 35** (Substitution). *We have $[\![t[x \leftarrow u]]\!]_\varphi = [\![t]\!]_{\varphi[x/[\![u]\!]_\varphi]}$.*

Proof. By induction on t. The cases for constants and variables are immediate. The case for applications follows from the induction hypotheses. The case for abstractions is similar to the case for families of propositions.

- Projections: We have $[\![(\pi_j\ F)[x \leftarrow u]]\!]_\varphi = [\![\pi_j\ F[x \leftarrow u]]\!]_\varphi = \mathsf{App}([\![F[x \leftarrow u]]\!]_\varphi, j)$. By induction hypothesis, we have $[\![F[x \leftarrow u]]\!]_\varphi = [\![F]\!]_{\varphi[x/[\![u]\!]_\varphi]}$. Therefore $[\![(\pi_j\ F)[x \leftarrow u]]\!]_\varphi = \mathsf{App}([\![F]\!]_{\varphi[x/[\![u]\!]_\varphi]}, j) = [\![\pi_j\ F]\!]_{\varphi[x/[\![u]\!]_\varphi]}$.
- Families: By definition, $[\![\{i \mapsto A_i\}_\mathcal{I}[x \leftarrow u]]\!]_\varphi$ and $[\![\{i \mapsto A_i\}_\mathcal{I}]\!]_{\varphi[[\![u]\!]_\varphi/x]}$ are respectively the unique members of $\mathsf{M}_{o_\mathcal{I}}$ such that, for any $j \in \mathcal{I}$, we have:

$$\mathsf{App}([\![\{i \mapsto A_i[x \leftarrow u]\}_\mathcal{I}]\!]_\varphi, j) = [\![A_j[x \leftarrow u]]\!]_\varphi$$
$$\mathsf{App}([\![\{i \mapsto A_i\}_\mathcal{I}]\!]_{\varphi[[\![u]\!]_\varphi/x]}, j) = [\![A_j]\!]_{\varphi[[\![u]\!]_\varphi/x]}$$

  By induction hypothesis, $[\![A_j[x \leftarrow u]]\!]_\varphi = [\![A_j]\!]_{\varphi[[\![u]\!]_\varphi/x]}$, which achieves the case. ◀

▶ **Lemma 36** (Model). $[\![_]\!]__$ *is a model.*

Proof. It only remains to show that $[\![t]\!]_\varphi = [\![u]\!]_\varphi$ when $t \equiv u$. We show it by induction on the derivation of $t \equiv u$.

- We have $[\![(\lambda x.\ t)\ u]\!]_\varphi = \mathsf{App}([\![\lambda x.\ t]\!]_\varphi, [\![u]\!]_\varphi) = [\![t]\!]_{\varphi[x/[\![u]\!]_\varphi]}$. Using Lemma 35, we derive $[\![(\lambda x.\ t)\ u]\!]_\varphi = [\![t[x \leftarrow u]]\!]_\varphi$.
- $[\![\lambda x.\ t\ x]\!]_\varphi$ is the unique member of its domain such that, for every $d \in \mathsf{M}_\tau$, we have $\mathsf{App}([\![\lambda x.\ t\ x]\!]_\varphi, d) = [\![t\ x]\!]_{\varphi[d/x]} = \mathsf{App}([\![t]\!]_{\varphi[d/x]}, [\![x]\!]_{\varphi[d/x]}) = \mathsf{App}([\![t]\!]_\varphi, d)$. Therefore $[\![\lambda x.\ t\ x]\!]_\varphi = [\![t]\!]_\varphi$.
- We directly have $[\![\pi_j\ \{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi = \mathsf{App}([\![\{i \mapsto A_i\}_\mathcal{I}]\!]_\varphi, j) = [\![A_j]\!]_\varphi$.
- We know that $[\![\{i \mapsto \pi_i\ F\}_\mathcal{I}]\!]_\varphi$ is the unique member of $\mathsf{M}_{o_\mathcal{I}}$ such that, for any $j \in \mathcal{I}$, we have $\mathsf{App}([\![\{i \mapsto \pi_i\ F\}_\mathcal{I}]\!]_\varphi, j) = [\![\pi_j\ F]\!]_\varphi = \mathsf{App}([\![F]\!]_\varphi, j)$. Thus $[\![\{i \mapsto \pi_i\ F\}_\mathcal{I}]\!]_\varphi = [\![F]\!]_\varphi$.
- The other cases follow from the induction hypotheses. ◀

The strong completeness result and the cut-elimination theorem are now easy to derive.

▶ **Theorem 37** (Strong Completeness). *If* $\omega([\![\bigwedge \Gamma]\!]_\varphi) \leq \omega([\![A]\!]_\varphi)$ *for any model, then* $\Gamma \vdash^\uparrow A$.

Proof. We use the interpretation in the Heyting applicative structure above with the assignment $\varphi = [x \leftarrow \langle [x], \rho(x) \rangle]$. We know that $[\![\bigwedge \Gamma]\!]_\varphi \in \mathsf{M}_o$. Its first member is $\|[\![\bigwedge \Gamma]\!]_\varphi\|_1 = [\bigwedge \Gamma]$ by Lemma 33 and by construction of φ. Its second member is $\omega([\![\bigwedge \Gamma]\!]_\varphi)$. Therefore $\lfloor \bigwedge \Gamma \rfloor \leq \omega([\![\bigwedge \Gamma]\!]_\varphi)$. Similarly, we have $\omega([\![A]\!]_\varphi) \leq \lceil A \rceil$. By hypothesis, we thus derive $\lfloor \bigwedge \Gamma \rfloor \leq \lceil A \rceil$. Since $\Gamma \in \lfloor \bigwedge \Gamma \rfloor$, we have $\Gamma \in \lceil A \rceil$. By definition of $\lceil A \rceil$, we conclude that $\Gamma \vdash^\uparrow A$. ◀

▶ **Corollary 38** (Cut Elimination). *If* $\Gamma \vdash A$ *then* $\Gamma \vdash^\uparrow A$.

Proof. Immediate combination of Theorems 20 and 37. ◀

5 Barr's Theorem for Higher-Order Geometric Theories

Geometric formulas are built using finite conjunctions, infinite disjunctions and existential quantifiers. Geometric implications are formulas of the form $\forall \vec{x}.(A \Rightarrow B)$, where A and B are geometric formulas and $\vec{x}$ is a finite sequence of variables (possibly empty). A theory $\mathbb{T}$ is geometric when all its axioms are geometric implications. Barr's theorem states that, in infinitary logic, any geometric implication that is classically provable in a geometric theory is also intuitionistically provable in the same theory. Note that the theorem refers to infinitary logic and not just geometric logic, because the principle of excluded middle cannot be expressed by a geometric formula or a geometric implication.

In this section, we extend Barr's theorem [2] to higher-order geometric theories. We follow the approach based on Friedman's translation [11], which has previously been used to prove Barr's theorem with finite disjunctions [28, 39, 38]. Throughout this section, we consider a higher-order infinitary logic with infinite disjunctions, but with finite conjunctions.

5.1 Friedman's Translation for Higher-Order Infinitary Logic

Double-negation translations [21, 15, 12, 22] embed classical logic into intuitionistic logic by applying the operator $\neg\neg A := (A \Rightarrow \bot) \Rightarrow \bot$ at various places in formulas – depending on the chosen translation. Friedman's trick [11] parameterizes these translations by a formula R. Instead of applying $\neg\neg A := (A \Rightarrow \bot) \Rightarrow \bot$, the parameterized translations apply the operator $TA := (A \Rightarrow R) \Rightarrow R$ and translate classical logic into intuitionistic logic. The parameterized version of Kuroda's double-negation translation can be extended to higher-order logic [38]. We adapt it so that it takes into account infinite disjunctions.

▶ **Definition 39** (Friedman's translation for higher-order infinitary logic). *Let A be a formula in higher-order infinitary logic. Its Friedman's translation is $A^T := TA_T$, where $t \mapsto t_T$ is inductively defined by:*

$$
\begin{array}{lcll}
x_T & := & x & \\
c_T & := & \begin{cases} \lambda p.\ \forall x.\ T(p\ x) & \text{if } c = \forall \\ \lambda p.\ \lambda q.\ p \Rightarrow Tq & \text{if } c = \Rightarrow \\ c & \text{otherwise} \end{cases} & \\
(\lambda x.\ t)_T & := & \lambda x.\ t_T & \\
(t\ u)_T & := & t_T\ u_T & \\
(\{i \mapsto A_i\}_{\mathcal{I}})_T & := & \{i \mapsto (A_i)_T\}_{\mathcal{I}} & \\
(\pi_j\ F)_T & := & \pi_j\ F_T &
\end{array}
$$

As we are in higher-order logic, our translation must commute with substitution and preserve convertibility. Note that all the results of this section also work if we only consider the β-reduction rule and the projection rules.

▶ **Proposition 40.** *Let t and u be terms and A and B be higher-order infintary formulas.*

1. $(t[z \leftarrow w])_T = t_T[z \leftarrow w_T]$.
2. $(A[z \leftarrow w])^T = A^T[z \leftarrow w_T]$.
3. *If $t \equiv u$ then $t_T \equiv u_T$.*
4. *If $A \equiv B$ then $A^T \equiv B^T$.*

Proof. The first item is proved by induction on the term t. The second item is a corollary of the first item. The third item is proved by induction on $t \equiv u$. The fourth item is a corollary of the third item. ◀

5.2 Translation from Classical Logic to Intuitionistic Logic

Let us remark that the operator $TA := (A \Rightarrow R) \Rightarrow R$ corresponds to a monad operator in the sense of [38], meaning that it satisfies the (unit) and (bind) properties.

$$\vdash_{\mathsf{L}} A \Rightarrow TA \qquad \text{(unit)}$$

$$\vdash_{\mathsf{L}} (A \Rightarrow TB) \Rightarrow (TA \Rightarrow TB) \qquad \text{(bind)}$$

It follows that the following properties hold [38].

▶ **Proposition 41.** *Let A and B be formulas and P be a predicate.*

1. $\vdash_{\mathsf{IL}} TTA \Leftrightarrow TA$
2. $\vdash_{\mathsf{IL}} (A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$
3. $\vdash_{\mathsf{IL}} T(A \wedge B) \Leftrightarrow (TA \wedge TB)$
4. $\vdash_{\mathsf{IL}} T(A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$
5. $\vdash_{\mathsf{IL}} T(A \Rightarrow TB) \Leftrightarrow (TA \Rightarrow TB)$
6. $\vdash_{\mathsf{IL}} T(\forall x.T(P\ x)) \Leftrightarrow \forall x.T(P\ x)$
7. $\vdash_{\mathsf{IL}} T(\exists x.T(P\ x)) \Leftrightarrow T(\exists\ P)$

We are now able to prove that the Kuroda-style Friedman's translation for higher-order logic with infinite disjunctions goes from classical logic to intuitionistic logic.

▶ **Theorem 42.** *Let $TA = (A \Rightarrow R) \Rightarrow R$ with parameter R. Let Γ, A be higher-order infinitary formulas. Let $\mathbb{T}$ be a higher-order geometric theory. If $\Gamma \vdash_{\mathsf{CL}} A$ in $\mathbb{T}$ then $\Gamma_T \vdash_{\mathsf{IL}} A^T$ in $\mathbb{T}$.*

Proof. We proceed by induction on the derivation. Most of the cases are direct consequences of Proposition 41 and are detailed in [38, see Lemma 16]. We only show the important cases.

- OrInf-I: By induction hypothesis, we have $\Gamma_T \vdash_{\mathsf{IL}} T(\pi_j\ F)_T$. Using Ax, we have $\Gamma_T, (\pi_j\ F)_T \vdash_{\mathsf{IL}} (\pi_j\ F)_T$. Using OrInf-I, we get $\Gamma_T, (\pi_j\ F)_T \vdash_{\mathsf{IL}} \bigvee_{\mathcal{I}} F$. Using Imp-I and Proposition 41(2), we derive $\Gamma_T \vdash_{\mathsf{IL}} T(\pi_j\ F)_T \Rightarrow T(\bigvee_{\mathcal{I}} F)$. We conclude using the induction hypothesis.
- OrInf-E: By induction hypotheses, we have $\Gamma_T \vdash_{\mathsf{IL}} T(\bigvee_{\mathcal{I}} F)$ and $\Gamma_T, (\pi_i\ F)_T \vdash_{\mathsf{IL}} TC_T$ for every $i \in \mathcal{I}$. By Ax, we have $\Gamma_T, \bigvee_{\mathcal{I}} F \vdash_{\mathsf{IL}} \bigvee_{\mathcal{I}} F$. By weakening of the induction hypotheses, we have $\Gamma_T, \bigvee_{\mathcal{I}} F, (\pi_i\ F)_T \vdash_{\mathsf{IL}} TC_T$ for every $i \in \mathcal{I}$. Therefore, we derive $\Gamma_T, \bigvee_{\mathcal{I}} F \vdash_{\mathsf{IL}} TC_T$ using OrInf-E. We get $\Gamma_T \vdash_{\mathsf{IL}} T(\bigvee_{\mathcal{I}} F) \Rightarrow TC_T$ using Imp-I and (bind). We conclude using Imp-E with the first induction hypothesis.
- PEM: We prove $\Gamma_T \vdash_{\mathsf{IL}} T(A \vee \neg A)_T$ using

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma_T, B \Rightarrow R, A_T, \bot \Rightarrow R \vdash_{\mathsf{IL}} A_T}\ \textsc{Ax}}{\Gamma_T, B \Rightarrow R, A_T, \bot \Rightarrow R \vdash_{\mathsf{IL}} B}\ \textsc{Or-IL}}{\Gamma_T, B \Rightarrow R, A_T, \bot \Rightarrow R \vdash_{\mathsf{IL}} R}\ \textsc{Imp-E with } B \Rightarrow R}{\Gamma_T, B \Rightarrow R \vdash_{\mathsf{IL}} A_T \Rightarrow ((\bot \Rightarrow R) \Rightarrow R)}\ \textsc{Imp-I} \times 2}{\dfrac{\Gamma_T, B \Rightarrow R \vdash_{\mathsf{IL}} B}{\Gamma_T, B \Rightarrow R \vdash_{\mathsf{IL}} R}\ \textsc{Imp-E with } B \Rightarrow R}\ \textsc{Or-IR}}{\Gamma_T \vdash_{\mathsf{IL}} ((A \vee (A \Rightarrow ((\bot \Rightarrow R) \Rightarrow R))) \Rightarrow R) \Rightarrow R}\ \textsc{Imp-I}$$

  where B is an abbreviation for the formula $A_T \vee (A_T \Rightarrow ((\bot \Rightarrow R) \Rightarrow R))$. ◀

▶ Remark 43. The And-I case crucially depends on $\vdash_{\mathsf{IL}} (TA \wedge TB) \Rightarrow T(A \wedge B)$. The proof of this property does not extend to the infinite case. This is why, in this section, we consider finite conjunctions and not infinite conjunctions.

5.3 Constructivisation for Geometric Theories

Geometric formulas do not contain implications or universal quantifiers, while the translation $t \mapsto t_T$ only inserts T after universal quantifiers and implications. It follows that for any geometric formula A, we directly have $A_T = A$. The same result does not hold for geometric implications $\forall \vec{x}.(A \Rightarrow B)$. But geometric implications imply their translation.

▶ **Lemma 44.** *Let A and B be higher-order geometric formulas, and $\vec{x}$ be a finite sequence of variables (possibly empty). We have $\vdash_{\mathsf{IL}} (\forall \vec{x}.(A \Rightarrow B)) \Rightarrow (\forall \vec{x}.(A \Rightarrow B))_T$.*

Proof. Since $(A \Rightarrow B)_T = A \Rightarrow TB$, we directly have $\vdash_{\mathsf{IL}} (A \Rightarrow B) \Rightarrow (A \Rightarrow B)_T$ using (unit) on B. If $\vec{x}$ is empty, this achieves the case. Suppose that $\vec{x} = x_1, \ldots, x_n$. We derive $\vdash_{\mathsf{IL}} (A \Rightarrow B) \Rightarrow T(A \Rightarrow B)_T$ using (unit) on $(A \Rightarrow B)_T$. Using All-I, we derive $\vdash_{\mathsf{IL}} \forall x_n.((A \Rightarrow B) \Rightarrow T(A \Rightarrow B)_T)$. Then we use the distributivity of $\forall$ over $\Rightarrow$ to derive $\vdash_{\mathsf{IL}} (\forall x_n.(A \Rightarrow B)) \Rightarrow (\forall x_n.T(A \Rightarrow B)_T)$, that is $\vdash_{\mathsf{IL}} (\forall x_n.(A \Rightarrow B)) \Rightarrow (\forall x_n.(A \Rightarrow B))_T$. We iterate this procedure to get $\vdash_{\mathsf{IL}} (\forall \vec{x}.(A \Rightarrow B)) \Rightarrow (\forall \vec{x}.(A \Rightarrow B))_T$. ◀

▶ **Theorem 45.** *Let $\mathbb{T}$ be a higher-order geometric theory, A and B be higher-order geometric formulas, and $\vec{x}$ be a finite sequence of variables (possibly empty). If $\vdash_{\mathsf{CL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathbb{T}$ then $\vdash_{\mathsf{IL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathbb{T}$.*

Proof. Suppose that $\vdash_{\mathsf{CL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathbb{T}$. Then there exists some geometric implications Θ of $\mathbb{T}$ such that $\Theta \vdash_{\mathsf{CL}} \forall \vec{x}.(A \Rightarrow B)$. Using All-E, we instantiate the bound variables $\vec{x}$ by fresh variables also named $\vec{x}$ (without loss of generality) and obtain $\Theta \vdash_{\mathsf{CL}} A \Rightarrow B$. Using Weak and Imp-E, we get $\Theta, A \vdash_{\mathsf{CL}} B$. Applying Theorem 42, we derive $\Theta_T, A_T \vdash_{\mathsf{IL}} B^T$, that is $\Theta_T, A \vdash_{\mathsf{IL}} (B \Rightarrow R) \Rightarrow R$. By Lemma 44, we derive $\Theta, A \vdash_{\mathsf{IL}} (B \Rightarrow R) \Rightarrow R$. Choosing $R := B$, we get $\Theta, A \vdash_{\mathsf{IL}} B$. Using Imp-I and All-I, we derive $\Theta \vdash_{\mathsf{IL}} \forall \vec{x}.(A \Rightarrow B)$. We conclude that $\vdash_{\mathsf{IL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathbb{T}$. ◀

6 Conclusion and Future Work

We have extended simple type theory with a type, a constructor and destructors for families of propositions indexed by an arbitrary countable set. In this setting, the β-reduction neither satisfies strong normalization nor confluence. However, every well-typed term has a head normal form. We used this type theory to define a higher-order infinitary logic. It features higher-order quantifiers as well as conjunctions and disjunctions indexed by an arbitrary countable set. We have defined a cut-free natural deduction system that precludes traditional cuts and commutative cuts.

We investigated the semantics of this higher-order infinitary logic. We have defined a sound notion of model based on Heyting algebras, and we have shown a strong notion of completeness based on V-complexes. These constructions allow us to derive the cut-elimination theorem.

Besides the elimination of cuts from proofs, we also studied the elimination of the principle of excluded middle. We have extended Barr's theorem to higher-order geometric theories. In other words, we are able to turn any classical proof of a higher-order geometric formula from a geometric theory into an intuitionistic proof of the same formula from the same theory.

Barr's theorem originates from topos theory, which can be used to give a semantics to many logics. In particular, first-order geometric logic can be interpreted in a geometric category [25], and higher-order logic can be interpreted in an elementary topos [18]. Grothendieck toposes are both complete and co-complete geometric categories and elementary toposes. We could thus investigate a semantics of higher-order infinitary logic in terms of Grothendieck toposes.

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